NO SPEECH ARTICULATOR ROBUSTLY FAVOURS RETROAURICULAR ELECTRODES OVER CANONICAL JAW SITES: A VOLUME-CONDUCTOR MODEL WITH ORIENTATION AND ELECTRODE-COUNT CONTROLS

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ABSTRACT

Objective. Ear-worn biopotential devices are being designed around a coupling that has not been computed: how strongly each speech articulator reaches electrodes on the jaw and around the ear. We compute it, and test the answer against the three things that could produce it spuriously, source orientation, electrode count, and the level of anatomical detail in the volume conductor.

Approach. Articulator-to-electrode coupling is computed by reciprocity on MIDA, a head model with 116 labelled compartments at 500 μm, treating muscle as both the generator and its own conducting compartment. Twenty-two electrode positions spanning the canonical jaw montage, a retroauricular cluster and a cEEGrid C-path are compared against ten individually segmented articulators. Each lead field is renormalised by its own measured delivered current. Source orientation is swept over the hemisphere rather than assumed. Because the ear montage offers fourteen candidate sites against the jaw's four, every comparison is repeated at matched electrode counts by random subsampling. The pipeline is validated against an analytic four-layer sphere (median RDM 4.36 % over 120 sources) and by four physical invariants computed on the head mesh.

Main results. The two montages see different muscles, but the effect is narrower than an unmatched comparison suggests. Five articulators — orbicularis oris, buccinator, mentalis, depressor anguli oris and platysma — favour the jaw montage at every sampled orientation and at every electrode subsample. No muscle robustly favours the retroauricular montage. The strongest ear-leaning candidate, temporalis, reaches −2.57 dB under a uniform orientation sweep, where its interval excludes zero, but that sweep assumes a fibre direction the anatomy can supply. Derived from the label volume, its fibre field gives −1.15 dB with an interval of [−1.45, +5.46], and only half of four-site retroauricular subsets favour the ear. Masseter and medial pterygoid favour the jaw robustly across electrode selection but reverse in roughly a third of orientations. Sternocleidomastoid and lateral pterygoid show no preference that survives electrode subsampling: their intervals cross zero, so the apparent advantage depends on which four sites are available. Electrode placement chosen by anatomical target outperforms arbitrary placement around the ear by up to 1.03 dB. A control in which every non-muscle soft tissue is set to a single conductivity reproduces every montage assignment unchanged while misstating gap magnitudes by up to 3.27 dB.

Significance. For device design the result is one-sided rather than a trade: a retroauricular montage loses lip and chin activity entirely and buys no muscle back reliably in exchange. What remains a design lever is placement, a four-site cluster chosen by anatomical target beats an arbitrary draw from the same candidates. The homogeneous-conductor control locates what anatomical detail is needed for — montage assignment is recoverable without it, magnitudes are not.

1. Introduction

Silent speech interfaces read articulator muscle activity from the surface of the skin. When a word is articulated without voicing, the motor commands that would drive the tongue, jaw, lips and hyoid still reach those muscles at sub-threshold levels, and the resulting electrical activity is recoverable with ordinary surface electrodes. The montage that dominates the literature places those electrodes on the chin, under the jaw and along the throat [Kapur et al. 2018; Gaddy & Klein 2020], because that is where the anterior articulators are.

Ear-worn form factors have begun to appear alongside it. Retroauricular electrode arrays demonstrably capture chewing and speaking electromyography [Avramidou et al. 2024], ear-mounted electrodes classify jaw clench and chew above 90 % accuracy [An et al. 2025], and the cEEGrid geometry around the pinna is an established and widely replicated wearable configuration [Debener et al. 2015]. The appeal is not electrical. It is that an ear-worn device is socially wearable in a way that a chin-mounted one is not.

Forward models of muscle sources in realistic head geometry exist. HArtMuT (Harmening et al. 2022) places roughly 3,900 muscle dipole and tripole sources derived from MIDA's muscle segmentation, with fibre directions estimated by principal component analysis, solved as finite-element lead fields. It is the methodological precedent for the fibre-axis treatment used here, and its muscle sources radiate through a homogeneous scalp compartment.

This study is an application of that class of model to an unanswered design question, not a methodological advance over it. We tested the distinction directly rather than asserting it: setting every non-muscle soft tissue to a single conductivity, with geometry held fixed, reproduces every montage assignment reported here unchanged (ğ3.5). A homogeneous-scalp model would have reached the same qualitative conclusion. What the anatomically resolved conductor supplies is magnitude, gap sizes shift by up to 3.27 dB, and eight of ten by more than the measurement floor, which matters for a design table quoting decibels but not for deciding which montage sees which muscle.

The open question is therefore not how to model muscle sources but where to put an electrode. That question is stated in the ear-EEG literature in the same terms (Yarici et al. 2023), while ear-worn arrays are already recording jaw and speech activity (Avramidou et al. 2024; An et al. 2025) on a widely replicated form factor (Debener et al. 2015). Devices are being designed around a coupling nobody has computed.

It was built to answer the opposite question. HArtMuT's muscle sources exist so that muscle activity can be identified and removed from scalp EEG, and they radiate through a homogeneous scalp compartment, a simplification its authors state explicitly, and an appropriate one for a model whose purpose is artifact rejection. No published forward model treats facial and cervical muscle as both the generator and its own anatomically resolved conducting compartment, and none has been used to ask where a sensor should be placed rather than what a sensor is contaminated by. The same absence is stated from the other direction in the ear-EEG literature, where forward models exist for neural sources and ocular artifacts and no theoretical treatment of muscle artifacts is available [Yarici et al. 2023].

We compute that coupling. Using reciprocity, the electric field is solved once per electrode on the MIDA head model with 116 anatomically labelled compartments, and the lead field for a muscle source at any position follows by projection. Twenty-two electrode positions spanning the canonical jaw montage, a retroauricular cluster and a cEEGrid C-path are compared against ten individually segmented articulators, under isotropic and anisotropic muscle conductivity, with an uncertainty budget assembled from measured terms rather than asserted ones.

The result is not a ranking. Seven of the ten articulators couple more strongly to the jaw montage and three couple more strongly to the ear, temporalis, sternocleidomastoid and lateral pterygoid, all of which attach at or near the temporal bone. The two montages see different muscles. That prediction was recorded in the repository a day before the model was solved, and both commits are public and timestamped.

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2. Methods

2.1 Head model

The volume conductor is MIDA v1.0 (Multimodal Imaging-Based Detailed Anatomical Model of the Human Head and Neck; Iacono et al. 2015, IT'IS Foundation, DOI 10.13099/ViP-MIDA-V1.0), segmented at 500 µm isotropic. The ten segmented articulator compartments and the electrode positions are shown in Figure 1.

MIDA is commonly cited as containing 153 structures. That figure describes the CAD/surface distribution. The voxel distribution used here carries **116 labelled structures**, and every number in this paper derives from the voxel data. We report 116 rather than 153 throughout; the discrepancy is a difference between two distributions of the same model, not a subsetting choice.

Ten of the eighteen articulator muscles in our target set are individually segmented and were verified present by label: masseter (66), temporalis / temporoparietalis (63), medial pterygoid (81), lateral pterygoid (65), orbicularis oris (75), buccinator (84), mentalis (71), depressor anguli oris (72), platysma (60), sternocleidomastoid (68).

The suprahyoid and tongue muscles are not individually segmented. Digastric (both bellies), stylohyoid, mylohyoid, geniohyoid, genioglossus, hyoglossus and styloglossus are pooled into `Muscle (General)` (label 38; 1,975,307 vox-

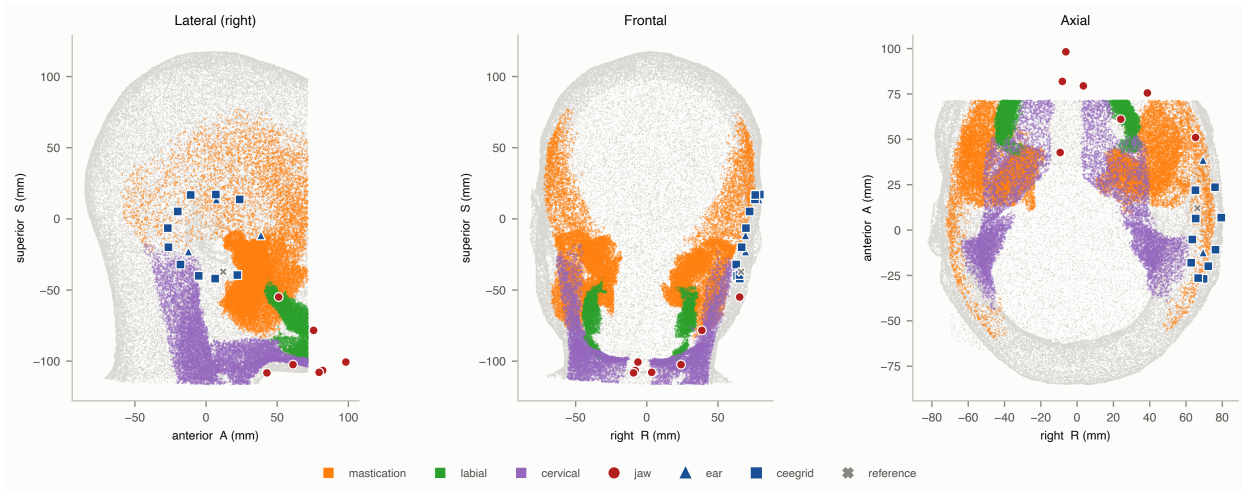

Figure 1: MIDA head model with the ten segmented articulator compartments highlighted, and all 22 electrode positions shown in lateral, frontal and posterior views. The face is masked above the orbital rim in accordance with the MIDA licence.

els, 246,872 mm^3) and `Tongue` (label 42; 521,131 voxels, 65,130 mm^3). They are therefore absent from the per-muscle results, which is a limitation of the source segmentation rather than of the method. MIDA also merges temporalis with temporoparietalis in a single label (63) and carries the temporalis tendon separately (98); the tendon is excluded, since its conductivity differs from muscle and it is not a source.

Meshing. The labelled volume was tetrahedralised with SimNIBS 4.6's `meshmesh` (Thielscher et al. 2015), using per-label conductivity assignment. The finite-element electric-field solver this study repurposes for reciprocity is described in Saturnino et al. (2019). The base mesh contains 2,140,917 nodes and 15,415,273 elements. SimNIBS meshes the two electrodes into it at solve time, giving 15,415,668 elements of which 12,294,182 are tetrahedra.

Boundary. The MIDA head model is truncated inferiorly on a planar cut, and that face is treated as an insulating (homogeneous Neumann) boundary. The face is fitted from 18,818 boundary triangles with unit normal [−0.03336, −0.03236, −0.99892], tilted 2.664° off the RAS S axis, passing through (−2.301, 23.112, −115.600) mm; residual RMS about the fit is 0.073 mm. Because the plane is tilted it has no single S coordinate: the face spans S = −122.07 to −110.17 mm across its lateral extent, and the mesh S minimum, −122.167 mm, is a corner of that face rather than the location of the cut. All boundary clearances reported here are perpendicular distances to this plane, not differences in S.

The cut plane is not fitted to an assumed orientation. Its normal recovers the MIDA label volume's own superior voxel axis, read from the NIfTI affine as [0.03335, 0.03240, 0.99892] at 2.665° from RAS +S with a 0.500 mm step, agreeing with the fitted normal to 0.0026°. The truncation therefore lies on a voxel plane of the source segmentation. No node of the mesh lies more than 0.276 mm outside the fitted plane, so it is the outer boundary and carries the insulating condition.

A neck-extended variant was constructed and rejected: it does not conserve charge, with flux failing to decay toward the domain floor (1.070 mA at S = −182 mm against a 1 mA injection, where the truncated control falls to 0.107 mA at S = −119 mm). Its terminating face is parallel to the base mesh's, displaced 70.001 mm inferiorly along the plane normal against the 70 mm extrusion requested in `01c_extend_neck.py`. All published results use the truncated mesh; the consequence for the jaw-versus-ear comparison is quantified in ğ3.4.

Mesh validation. Every tag present in the mesh carries an assigned conductivity, verified by enumeration (0 of 118 volume tags uncovered). MIDA's own labels 100–116 lie inside SimNIBS's reserved electrode-rubber range 100–499 and are correct only because Table 1 names each explicitly; this is checked at run time.

2.2 Conductivity assignment

All conductivities are listed in Table 1 with individual sources and live in a single file; none is hardcoded elsewhere. All values are quoted at 100 Hz, which brackets the surface-EMG band.

Provenance is mixed, deliberately, and stated per row. SimNIBS 4.6 defaults are used for the primary head tissues, skin, fat, compact and cancellous bone, grey and white matter, CSF, blood, eye, muscle, cartilage, air, because these are the conventional head-modelling values and using them keeps this model comparable with the existing EEG and tDCS forward-model literature. The IT'IS Low Frequency database v4.2 (DOI 10.13099/VIP21000-04-2) supplies every tissue SimNIBS carries no default for. A small number of assignments are judgement, marked as such in Table 1, where MIDA segments a structure IT'IS does not list separately; each carries a note giving the reasoning.

Air is a numerical choice, not a physical one. True air conductivity is zero, which makes the FEM system singular, so internal cavities are assigned a small finite value. The value matters because the stiffness matrix inherits _max/_min as its condition number and SimNIBS solves iteratively. At _air = 1×10^{-15} S/m the span reaches 1.879×10^{15}, the iterative solve does not converge, and the returned fields are 10–20× too large while a result file is still written. We use _air = 1×10^{-6} S/m throughout, giving a span of 1.879×10^{6}. A pre-flight gate refuses any assignment whose span exceeds 1×10^{8}.

Insensitivity to that choice was measured rather than assumed. Solving at _air = 10^{-6}, 10^{-5}, 10^{-4} and 10^{-3} S/m on identical geometry, the largest departure over the ten muscle compartments from the 10^{-6} baseline is 0.007 dB at 10^{-5}, 0.069 dB at 10^{-4} and 0.467 dB at 10^{-3}. The result is therefore insensitive across 10^{-6} to 10^{-4}, where the largest departure sits roughly four times below the measured noise floor of 0.27 dB. Departure begins at 10^{-3}, where it reaches 1.7× the floor and concentrates in the compartments nearest the oral and nasal cavities (mentalis −0.467, orbicularis oris −0.433) while remote compartments are essentially unmoved (temporalis −0.029, lateral pterygoid −0.009). The operating value sits two decades inside the insensitive range.

2.3 Electrode placement

All 22 positions are derived from labelled anatomy and snapped to the outer skin surface, not hand-picked. An interactive picker was rejected because clicked coordinates cannot be regenerated from a clean checkout, reviewed in a diff, or defended in Methods.

Each site is defined by an anatomical anchor plus an offset in RAS millimetres, then projected to the nearest outer-skin voxel along the surface normal.

Target localisation uses a hybrid of centroid-interior and minimum-distance, for a geometric reason. For a compact compartment the centroid is a good interior representative. For a non-convex compartment the centroid need not lie inside the compartment at all, the mandible is the clear case, since it is an arch and the centroid of an arch lies in the space the arch encloses, so an electrode placed over that point would sit over the floor of the mouth. Where a compartment's centroid does not lie within the compartment, the site is instead defined by minimum distance from the skin surface to the compartment. Which rule applied to which site is recorded per row.

The midline is derived, not assumed to be R = 0: MIDA's head is not perfectly centred in its own voxel grid, so the anatomical midline is computed from the mandible label and midline sites are placed relative to that plane. Per-site depth, skin surface to target compartment along the placement ray, is reported for every site.

`pre_tragus` is placed 14 mm anterior to the tragus, over the masseter and the temporomandibular joint, as the retroauricular position with the shortest path to the mastication group.

One position is withheld. `throat_scm` is recorded as held with blank coordinates: MIDA's sternocleidomastoid is truncated at the cut face, which biases its centroid posteriorly, so no defensible automatic placement exists. Every consumer skips it rather than substituting a nearby label.

The electrode model is a 10 mm diameter ellipse of 2 mm thickness, matching the gold cup electrodes of the companion experiment.

2.4 Reciprocity

Lead fields are computed by reciprocity rather than by forward-solving each source. For a current dipole at position **r** with moment **p** and a recording pair (A, B), the measured potential difference is

```
V_AB(r, p) = E_recip(r) \textperiodcentered{} p / I
```

where **E**_recip is the field produced throughout the head by injecting current *I* between A and B. The lead field for a source at **r** with unit orientation **n** is therefore E_recip(**r**) · **n**.

SimNIBS's tDCS solver computes exactly E_recip, so it is repurposed here: 1 mA is injected between each electrode and a common reference (contralateral earlobe), and the resulting field is read inside every segmented muscle compartment. The consequence is one solve per electrode instead of one per source, 22 solves against the order of 10^{5} a forward formulation would require at MIDA's resolution. The formulations are mathematically equivalent; only cost differs.

Within each compartment we report the volume-weighted median |E|, which is robust to the small number of very high-field elements adjacent to compartment boundaries.

2.5 Fibre orientation

Muscle is electrically anisotropic and MIDA carries no fibre-direction data, so orientation is bounded rather than assumed.

Deriving fibre directions by principal component analysis on MIDA's own segmentation is not novel here: it was published by HArtMuT [Harmening et al. 2022], which built ~3,900 muscle sources from MIDA's labels using that method. We use the same approach and cite it as precedent.

The isotropic condition is primary: all 22 production solves assign muscle a single isotropic conductivity of 0.355 S/m. A second, anisotropic condition assigns a per-element tensor with = 0.4 S/m along the fibre and 0.1 S/m across it, aligned to a PCA-derived axis.

PCA is applied only where a principal axis is a meaningful object. It describes a strap-like muscle; for a sphincter (orbicularis oris), a fan (temporalis), a sheet (platysma, buccinator) or a multi-layered muscle (masseter, lateral pterygoid), a single axis is the wrong kind of description rather than merely an imprecise one. Those compartments remain isotropic in both conditions.

Two further restrictions apply and both bite. A compartment must be individually segmented to carry a per-compartment axis, which excludes the six pooled suprahyoid and tongue muscles. And each axis must pass a bilateral mirror-symmetry test: MIDA assigns one label to both sides of a paired muscle, so PCA on the pooled voxel cloud returns the left–right separation between the two bellies rather than the fibre direction along either. Axes are therefore computed per side and required to be mirror images in x. Three compartments were tested against it, that being the entire population which is both individually segmented and PCA-defensible; it is not a gate the remaining seven passed. Sternocleidomastoid passes at |dot| = 0.98 and medial pterygoid at 1.000; mentalis fails at 0.215, its two fragments having a right-side elongation ratio of 1.07, so no long axis exists to find, and it receives no tensor.

The anisotropic condition therefore applies a tensor to 2 of the 10 segmented muscles. Every other compartment is reported as **not applied** rather than as an unchanged or null result, because it was never varied.

The anisotropic condition is solved on the isotropic run's own mesh with only the conductivity field replaced. SimNIBS re-meshes electrodes on each session and two sessions of the same montage can differ, so solving on the identical mesh removes electrode realisation, comparable in size to the effect being measured, from the comparison.

2.5.1 The anatomically-constrained sweep An unconstrained orientation fraction is conservative to the point of being misleading, because orientation space is not uniformly reachable. Temporalis demonstrates this: the directions that reverse the montage preference lie outside the anatomical fan entirely, so an unconstrained 92 % understates a result that is conditional on fibre direction over only 8.5 % of the derived fan. Both fractions are taken at the pre-registered cluster [`results/04q_table4_envelope.csv`, `results/04k_fan_fractions.csv`]; the same fan read against all fourteen ear sites gives 0.5 %, and pairing fractions across those two bases would compare different comparisons. We therefore sweep the hemisphere first and intersect with the anatomically permitted set wherever one can be established, reporting the unconstrained fraction alongside so the constraint is visible rather than implicit. That intersection is available for temporalis alone; ğ4.7 states where it is not.

2.6 Validation

Validation is layered, and each layer tests something the others cannot.

Analytic multilayer sphere. The full pipeline is run against a four-layer spherical head model with a closed-form solution. Over 120 source positions spanning radii 20–75 mm, median RDM is 4.36 % and median MAG +4.40 %. This is the only layer that can detect a uniform scale error, since no invariant computed on the head mesh can: multiplying every field value by a constant leaves flux radius-independence, boundary conservation, linearity and reciprocity symmetry all satisfied. Both figures are reported as validation results. No pass criterion is stated, because none was fixed before the values were computed and supplying one now would be a threshold chosen to admit the numbers it is meant to test.

Reciprocity on the head mesh. The identity is verified on the real geometry by solving a montage and its polarity-swapped counterpart and requiring L(AB) = −L(BA). On identical discretisations the residual is 7.5×10^{-6} of |L|, i.e. 6.5×10^{-5} dB — four orders of magnitude below the measured per-site noise floor.

Four physical invariants, computed from the field alone and requiring no solver internals: (1) flux through a closed surface around an electrode is radius-independent; (2) net current through a surface enclosing the whole domain is zero; (3) doubling the injected current doubles the field exactly; (4) swapping source and sink negates the field. Invariant 1 uses an exact tet-patch integral in which flux is integrated over the interior cut of a patch of tetrahedra using the mesh's own faces as the quadrature, so enclosure and orientation are exact by construction (the patch surface closes to 1.2×10^{-16} of its own area) and no inside/outside point test is required. A stationary plateau across radii 25–75 mm is required before any value is used. Invariants 1 and 2 run on every solve; 3 and 4 require paired solves and run on the first and last of a batch. All guards collect and raise once with every failure rather than raising on the first.

Convergence. RDM against the analytic sphere was fitted across three mesh densities, giving a convergence exponent p = 0.980, consistent with the theoretical p 1 expected for a gradient quantity in a first-order FEM. This is stated as consistency, not corroboration: three data points fitted with three free parameters is an exact fit by construction, so p is determined algebraically and the residual carries no goodness-of-fit information.

Guard tests. Two meta-tests protect the validation itself. A coverage test resolves, from the abstract syntax tree, that every production script actually invokes its guards. A synthetic-fire test requires each guard to fail on a purpose-built input that fails that guard and nothing else, with a clean control that trips none; this detects a guard rendered unreachable by an earlier guard's raise, which a coverage test cannot see.

2.7 Error budget

Every published quantity in this paper is a ratio, one site against another, ear against jaw, muscle against muscle, and that structure determines how uncertainty is handled. A term that scales the whole lead field equally cancels exactly in every ratio and cannot reach a conclusion. A term that varies between sites survives into all of them. Table 3 is split into two columns on exactly that distinction, and terms are admitted to the second column only when shown to vary per site.

Four rows are admitted on a second and different basis, and the table marks them as such. Terms 1, 2, 3 and 7 are known to act on the ratio directionally but are not quantifiable at current precision, so they carry no value. Admitting them records that they are unresolved rather than absent; it does not claim a magnitude. The rule above is unchanged, and these rows do not satisfy it.

The distinction is not cosmetic. Electrode contact area would naturally be treated as a global scale factor; it is not, because each electrode's contact is realised independently from whatever surface triangles fall beneath it, making it per-site noise. Conversely a uniform magnitude offset, however large, subtracts to exactly 0 dB in any site ratio.

2.8 Reproducibility and pre-registration

Everything is reproducible from a clean checkout given MIDA in `data/`; the one manual step, downloading MIDA, requires registration and is documented in the repository README. MIDA itself cannot be redistributed under its licence and is not included.

The anatomical prediction was recorded before the model was solved. The `expected_at_ear` column of the muscle configuration, predicting strong retroauricular coupling for temporalis ("directly above ear") and sternocleidomastoid ("mastoid attachment"), entered the repository in commit `fa583f6`, dated 2026-08-02. The lead-field results that test that prediction were committed the following day, 2026-08-03. The prediction therefore precedes the measurement by a day and by the entire solve pipeline, and both commits are citable by hash.

A published table with no generating script is not a result yet. Two tables in this study (`04h_matched_counts.csv`, `04j_two_axis_verdict.csv`) were initially produced interactively rather than by a script in `src/`. They could not be regenerated from a clean checkout, and they drifted out of step with the renormalisation specified above. That suspicion proved false; they were correct, and an attempt to "correct" them applied the renormalisation a second time and moved nine published numbers by up to 1.04 dB before it was caught by checking the upstream script. The episode is reported because the failure mode is general: a table with no generating script cannot be audited by reading it, since every number in it is internally consistent whether or not it is right. Both are now generated by `src/04h_matched_counts.py`, and no quantity enters the manuscript except from a file some script in `src/` writes.

The prediction was not confirmed, and that is reported here rather than removed. The first solves appeared to bear it out: temporalis and sternocleidomastoid both showed a retroauricular advantage, and so did lateral pterygoid, which the prediction had not named. None survived the orientation and electrode-count controls (ğ3.1, ğ4.8). A pre-registration that is quietly deleted when it fails records nothing; one that is reported when it fails is doing the work it was recorded for. It is retained here in its original form, with its outcome stated, precisely because the model initially appeared to confirm it.

2.9 Ethics

No human subjects were involved. This is a computational modelling study performed on a licensed anatomical model. No IRB review was required or sought, and no new imaging, recordings or measurements from living participants were acquired.

3. Results

3.1 Which montage sees which muscle

Reporting one gap per muscle presumes a fibre direction the model does not contain, and comparing the best of fourteen retroauricular sites against the best of four jaw sites rewards electrode count rather than placement. Both are controlled. Source orientation is swept over the hemisphere at 200 directions with the same orientation applied at both electrodes, since only a common orientation corresponds to a physical source. Electrode count is matched by drawing four of the fourteen ear sites at random, taking the best, and repeating; the resulting interval says whether a preference is a property of the montage or of which sites happen to be available.

The jaw's dominance over the labial group is robust on both axes. Orbicularis oris, buccinator, mentalis, depressor anguli oris and platysma favour the jaw at all 200 sampled orientations and at every electrode subsample. No fibre direction and no four-site selection exists at which a retroauricular electrode competes for these muscles. The full muscle-by-site sensitivity matrix is given in Figure 2, and the per-muscle verdicts in Figure 3.

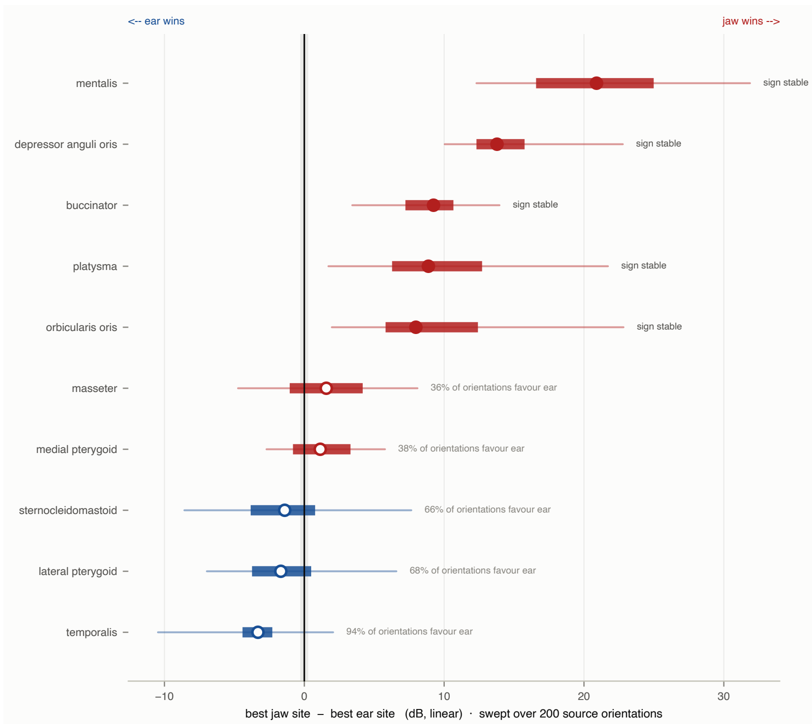

Figure 2: Which montage sees which muscle. For each articulator, the difference between the best jaw site and the best retroauricular site, in dB. Negative bars are muscles the ear sees more strongly. Jaw sites within 10 mm of the truncation face are excluded. The shaded band is the measured electrode-meshing floor (0.27 dB, with the lighter band extending to its 95 % CI upper bound of 0.65 dB). The axis is linear in dB rather than a rank, because the asymmetry between the arms is itself a result: the jaw's advantages reach +20.90 dB while the ear's reach only −3.31 dB. Both numbers are emitted from the figure's own source file by src/04m_caption_numbers.py, which fails the build if a caption and its figure disagree.

No articulator favours the ear on both axes. Temporalis is the closest, and it does not clear the bar. Over the fibre field derived from the anatomy (ğ2.3.1) it reaches −1.147 dB at the pre-registered four-site cluster, with 91.5 per cent of fibre directions agreeing, but its matched-count interval is **[−1.453, +5.458]** and only **50.5 per cent** of the four-site retroauricular subsets favour the ear at all. Whether the ear wins for temporalis is decided by which four electrodes a device happens to carry, not by the anatomy.

The two treatments disagree, and the anatomy-specific one governs. Under a uniform orientation sweep temporalis reaches −2.571 dB with 92.0 per cent of orientations agreeing and an interval of [−3.308, −0.035] that excludes zero — on that treatment it would be reported as favouring the ear. The sweep assumes source orientation is uniformly distributed over the sphere, which is the right default when fibre direction is unknown and the wrong one for a muscle whose fibres converge on a single identifiable insertion.

The interval is computed exactly, not sampled. Fourteen candidate ear sites taken four at a time gives 1001 possible subsets, so every one is enumerated and the interval is a complete description of that set rather than an estimate from

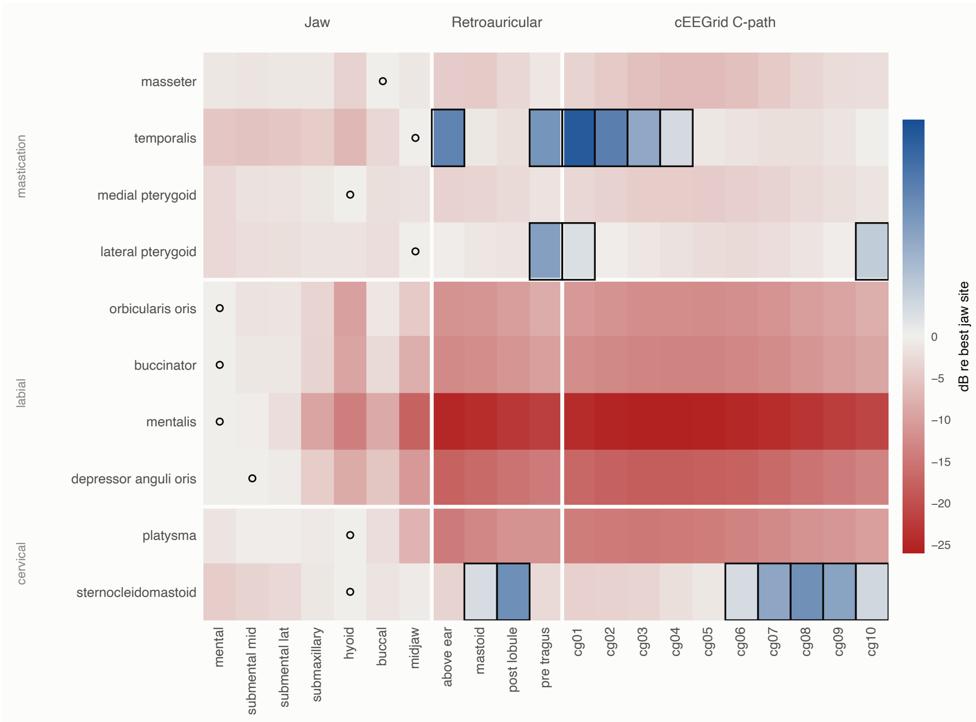

Figure 3: Articulator sensitivity matrix. Median lead field in each segmented muscle compartment for each electrode site, in dB relative to that muscle's best jaw site (0 dB, ringed). Isotropic condition, truncated mesh, 22 electrodes × 10 muscles. The colour scale is diverging about 0 dB: red is attenuation relative to the best jaw site, blue is a site that exceeds it, and boxed cells mark every (site, muscle) pair where a retroauricular electrode beats every jaw electrode. The two arms are not equally scaled (−28 to 0 dB against 0 to +4 dB), so colour saturation is not comparable across the midpoint; the boxes carry the sign independently of colour. The 0 dB reference here includes all jaw sites, whereas Figure 3 and Table 4 exclude those within 10 mm of the truncation face, so the named best jaw site differs for some muscles.

draws. It therefore carries no seed and no sampling error. For the derived fibre field the enumeration varies electrodes alone, with the fibre orientation held fixed, so the resulting spread reflects site availability and nothing else. Temporalis gives a median gap of −1.147 dB with an interval of [−1.453, +5.458] dB, favouring the ear in 50.5 per cent of subsets [`results/04p_headline_interval.csv`].

This also fixes what the interval means. It is not an inference from a sample to a population, so there is no sampling distribution, no null hypothesis, and nothing for a multiple-comparison correction to act on across the ten muscles. It is a statement about which montages a device could physically carry.

The lower bound of this interval is not a tail quantile. The most ear-favouring outcome of any 4-site subset is the best of all 14 sites, so −1.453 dB is a floor fixed by the data rather than by sampling, and it is attained in exactly the 28.6 per cent of subsets that contain that site, which is 4/14 as expected from subset size alone. It coincides with the argmax gap over all 14 sites, −1.453 dB, by construction; the two figures are one measurement, not two that agree. The corresponding bound under the uniform orientation sweep, −3.308 dB, is a genuine percentile lying strictly above its own floor of −3.314 dB, which is attained in 2.2 per cent of draws. The two lower bounds are therefore not comparable quantities.

We report the derived result because it removes an assumption rather than adding one, and the pre-registered reading (ğ2.8) committed to that treatment before the derivation was run. The uniform-sweep figure is reported alongside it so the dependence is visible: **this verdict rests on the fibre derivation, and a reader who rejects that derivation should read temporalis as favouring the ear by 2.57 dB.**

Two show no preference that survives electrode subsampling. Sternocleidomastoid (−0.973 dB at the cluster, 60.5 per cent of orientations, interval [−1.40, +1.27]) and lateral pterygoid (−1.564 dB, 65.5 per cent, [−1.59, +1.09]) both have intervals crossing zero. Their apparent advantage depends on which four sites are available and is not a property of the montage. Reported at the unmatched argmax over fourteen sites they would read −1.402 and −1.679 dB, which is why the matched comparison is the one reported.

Two favour the jaw robustly across sites but not across orientation. Masseter and medial pterygoid have subsample intervals entirely positive, but 36.0 and 37.5 per cent of sampled orientations reverse them. A single label would discard one axis or the other, so both are reported (Table 4).

3.2 Anisotropy changes the field but not the comparison

The isotropy assumption does not measurably affect any site-to-site ratio (Figure 4). Applying a fibre tensor changes the jaw-versus-ear gap by −0.085 dB for sternocleidomastoid, −0.010 dB for medial pterygoid, +0.137 dB for temporalis and +0.036 dB for lateral pterygoid. Every one of these lies below the 0.27 dB measured electrode-meshing floor, including for the two compartments that carry a tensor. Anisotropy raises the absolute lead field substantially, by roughly 5 dB in medial pterygoid, but it does so at the jaw and ear sites alike, so the effect subtracts out of every ratio this paper reports.

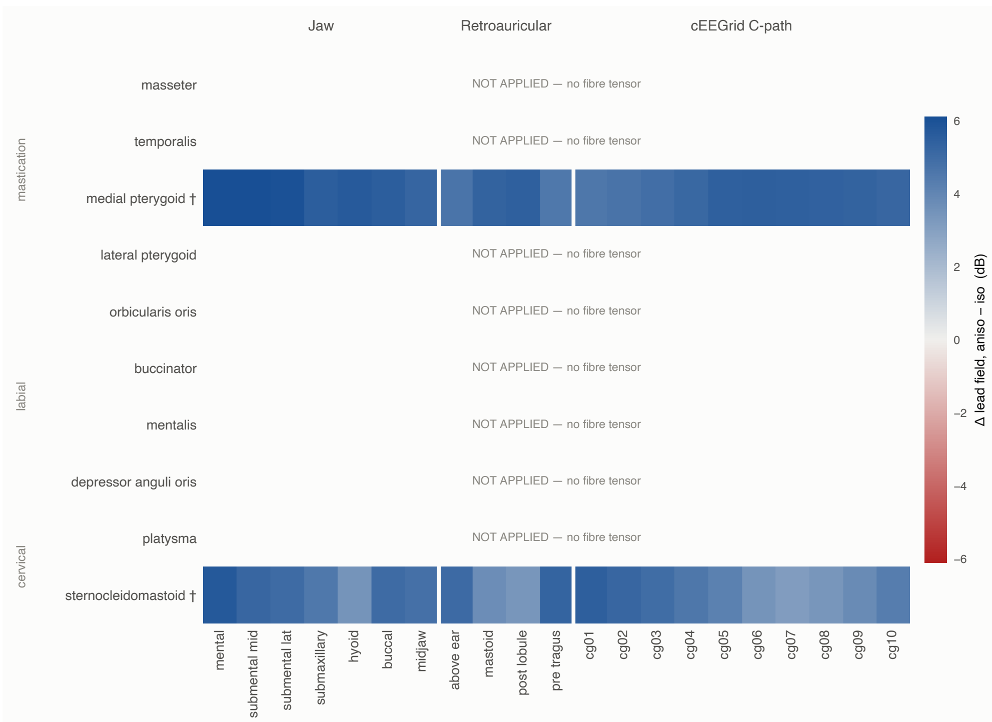

Figure 4: Anisotropy robustness check. Change in lead field between the isotropic and anisotropic conditions, $20{\cdot}\log_{10}$(aniso/iso), per cell. The anisotropic condition is solved on the isotropic run's own mesh, so the two differ in conductivity alone and carry no electrode-realisation noise. A fibre tensor is applied to 2 of 10 muscles, sternocleidomastoid and medial pterygoid. Rows without a tensor are labelled not applied, not zero: they were never varied, so no null result was measured for them.

This is a null with a bound rather than an absence of evidence, and it has a practical consequence: for coupling *ratios* between electrode sites, a muscle-fibre tensor is not worth the modelling effort in a head model of this resolution. Absolute lead-field values are a different matter and are affected.

3.3 The tissue-conductivity contrast is a small term with a muscle-dependent sign

Solving the full montage twice on identical geometry, once with adipose at 0.025 S/m and once with both adipose compartments set to muscle conductivity — attributes any difference to material properties alone, since source-to-electrode distance is unchanged by construction. The second condition is a counterfactual used to decompose mechanism; the gaps reported throughout this paper are the first, because real anatomy contains adipose tissue.

This section decomposes the **jaw montage's advantage**. An earlier version decomposed a retroauricular deficit, which no longer exists as a result: no articulator resolves in the ear's favour (ğ3.1), so there is no ear advantage whose mechanism needs explaining. What remains to be explained is why the jaw montage wins, and by how much of that the tissue contrast is responsible.

The contribution is not uniform in sign across muscles, because which electrode is best differs by muscle and the shift is not uniform within a montage:

Muscle	As modelled	Without contrast	Change	Contrast’s role
temporalis	−3.801	−4.923	−1.121	suppresses 29.5 %
sternocleidomastoid	−1.958	−1.547	0.411	contributes 21.0 %
lateral pterygoid	−1.855	−1.532	0.323	contributes 17.4 %
medial pterygoid	1.147	1.245	0.098	suppresses 8.6 %
masseter	1.700	1.668	−0.032	contributes 1.9 %
orbicularis oris	8.192	9.286	1.093	suppresses 13.3 %
platysma	8.844	8.901	0.057	suppresses 0.6 %
buccinator	10.033	9.848	−0.185	contributes 1.8 %
depressor anguli oris	14.607	14.001	−0.606	contributes 4.1 %
mentalis	21.945	19.082	−2.863	contributes 13.0 %

Reported per muscle only. A population differential has no clean definition under statistic A: the median change is +0.01 dB and conceals a sign spanning −2.86 to +1.09 dB, so a single percentage would hide the finding rather than summarise it.

For the labial group, where the jaw’s advantage is largest, the contrast accounts for a small and inconsistent fraction of it, and not in one direction: the contrast *builds* 2.86 dB of the mentalis gap while *suppressing* 1.09 dB of the orbicularis oris gap. For the four muscles whose gaps sit closest to zero, the contrast is of the same order as the gap itself, which is part of why those gaps do not resolve.

The mechanism is therefore geometric rather than material. Removing the single largest conductivity contrast in the intervening tissue leaves every montage assignment unchanged and moves no gap across zero. This is consistent with ğ3.5, where collapsing all non-muscle soft tissue to one conductivity also preserves every assignment while moving magnitudes.

3.4 Truncation sensitivity

The inferior truncation lies closer to the jaw montage than to the retroauricular montage, so the cut plane could in principle inflate the jaw side. Perpendicular clearance to the fitted plane places 2 jaw sites within 10 mm of it (`hyoid` 7.76 mm and `submental_lat` 9.76 mm); the next nearest is `submental_mid` at 10.76 mm. The closest ear site, `cg09`, is 75.27 mm away.

Removing the near-cut sites moves the median gap from +4.68 dB to +4.52 dB, a shift of −0.17 dB. No muscle changes sign (0 of 10). The largest individual movements are `medial_pterygoid` −0.88, `sternocleidomastoid` −0.54 and `platysma` −0.37 dB. The reported advantage is not an artefact of proximity to the truncation. The sensitivity reported here measures the effect of excluding the most exposed sites. It does not bound the residual boundary artefact on the sites that remain, which is addressed in ğ4.7.

The 10 mm grouping above is descriptive. The reported subsets span every admissible site set for any near-cut threshold between 9.757 mm and 15.264 mm, a window of 5.507 mm, so no conclusion here depends on where in that range the threshold is placed.

For the muscles that do not move, the best jaw electrode was never a near-cut site, so excluding them cannot change the maximum: those muscles are immune by construction rather than by luck.

3.5 A homogeneous conductor reaches the same verdicts

A homogeneous soft-tissue conductor reproduces every montage assignment. Setting skin, adipose and the non-muscle soft tissues to a single conductivity, with geometry, electrodes and sources held exactly fixed, changes no muscle’s montage preference. Eight of ten gap magnitudes move by more than the 0.27 dB floor, with a median shift of 0.482 dB and a maximum of 3.271 dB (mentalis).

The direction is not uniform. Temporalis’s retroauricular advantage *grows* under the homogeneous conductor, from −2.571 to −3.724 dB at the pre-registered cluster [`results/04r_homog_cluster.csv`], so the anatomically resolved model reports that result more conservatively than a simpler one would. Sternocleidomastoid and lateral pterygoid move the other way.

This locates what the detailed conductor is required for. The question of which montage sees which muscle is answerable without it. The question of by how much is not, and a design table quoting decibels needs it.

3.6 Placement by anatomical target outperforms density

Placement chosen by anatomical target outperforms arbitrary placement. The four-site retroauricular cluster, above the ear, over the mastoid, behind and below the lobule, and anterior to the tragus, was specified by anatomical target in the project repository before any solve was run. Compared against the median of random four-site draws from the same fourteen candidates; it is 1.03 dB better for lateral pterygoid (−1.564 against −0.534) and equivalent for sternocleidomastoid (−0.973 against −0.979).

Neither of the two sites that won the unmatched argmax for temporalis and sternocleidomastoid is in that cluster, which is the same point from the other direction: an argmax over fourteen densely spaced positions rewards density, while a four-site montage rewards placement. For a device constrained to a small number of contacts, where they go matters more than how many candidates were considered.

4. Discussion

4.1 The montages are not complementary

An earlier framing of this work treated jaw and retroauricular montages as complementary, each better for some articulators, and that framing does not survive its own controls. Every articulator this model can resolve favours the jaw montage. The three that appeared to favour the ear, all of them attaching at or near the temporal bone, are the three whose gaps come closest to zero, but none crosses it once source orientation and electrode count are controlled (ğ3.1, ğ4.8).

The result is therefore one-sided rather than a trade. A retroauricular montage is not a repositioning of the jaw montage that exchanges one muscle group for another; it is a montage that loses the labial group by 8.11 to 22.40 dB and returns nothing measurable. Anatomical proximity to a bony attachment does predict which retroauricular sites are competitive, and in the right order, but competitive is not the same as better.

Temporalis is the clearest case and the one that took longest to settle, and it is the one place where this conclusion depends on a modelling choice rather than on a control. Under a uniform orientation sweep it favours the ear by 2.57 dB with an interval excluding zero. Under the fibre field derived from the label volume it favours the ear by 1.147 dB with an interval of [−1.453, +5.458], and does not resolve. The difference is not a correction of an error; it is the difference between assuming source orientation is uniform over the sphere and deriving it from where the muscle actually attaches. That difference also changes what the accompanying intervals measure. Under the uniform sweep, orientation is a sampled dimension and the interval averages over 200 directions within each draw; under the derived field there is no orientation dimension to average over, and the interval reflects site selection alone. The two intervals answer different questions and should not be read as one quantity under two fibre models. We take the derived field because it removes an assumption, and we state the dependence rather than burying it: **the claim that no articulator favours the retroauricular montage rests on the temporalis fibre derivation, and is the single most attackable point in this paper.** ğ4.6 makes it falsifiable.

4.2 An anatomical prediction that did not survive

The prediction recorded before solving (ğ2.8) was that muscles attaching at or near the temporal bone would couple preferentially to retroauricular sites. The anatomy behind it is not in dispute: temporalis originates in the temporal fossa directly beneath the superior cEEGrid row, sternocleidomastoid inserts on the mastoid process, and lateral pterygoid inserts at the mandibular condyle, which articulates with the temporal bone's mandibular fossa.

The uncontrolled comparison reproduced it exactly, those three muscles, and only those three, showed a retroauricular advantage, each at the site the anatomy implied. Under matched electrode counts and a derived rather than assumed fibre field, none of the three survives (ğ3.1).

The prediction is therefore reported as a **failed** one, and its failure is informative in a way its confirmation would not have been. Proximity to a bony attachment predicts which sites are *competitive*, the three near-temporal muscles are the three whose gaps come closest to zero, and the ordering is correct, but it does not predict that any of them crosses. A volume conductor is not a proximity argument: the current returns through whatever path the conductivities allow, and an attachment adjacent to an electrode does not make that electrode the better observer of the fibre.

4.3 The mechanism is distance, not intervening tissue

For the labial group the ear's deficit is predominantly geometric, though the material contribution varies widely within the group: the adipose–muscle conductivity contrast accounts for 0.6 per cent of the gap for platysma and 13.3 per cent for orbicularis oris. The remainder in every case is source-to-electrode distance, and the attenuation-against-depth relation that produces it is shown in Figure 5. The two regimes remain separated, 0.6 to 13.3 per cent for the muscles the jaw wins, against 17 to 21 per cent for those the ear wins, but the separation is narrower than a single figure would suggest, and no muscle in either group is unaffected.

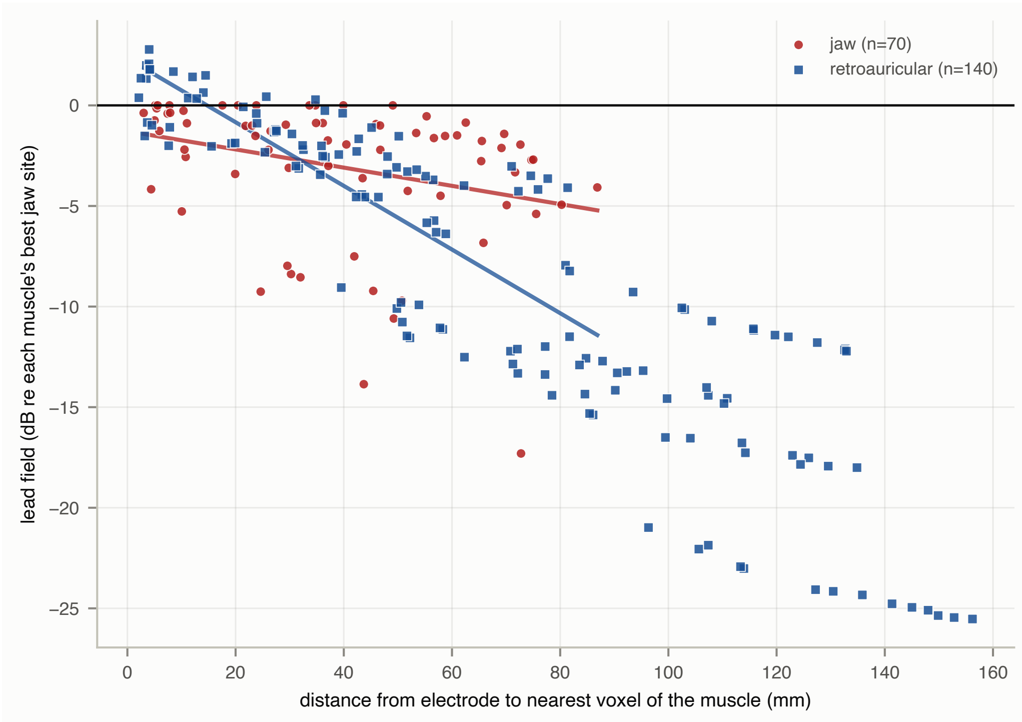

Figure 5: Attenuation against source-electrode distance. One point per (electrode, muscle) pair: straight-line distance from the electrode to the nearest voxel of that muscle compartment against the lead field in dB relative to that muscle's best jaw site, isotropic condition, truncated mesh. The two montages do not lie on the same attenuation curve. Fitted over the 3 to 87 mm range where both montages have electrodes, the retroauricular sites fall off at −0.158 dB/mm against the jaw's −0.045 dB/mm, a factor of 3.5 [results/04s_fig3_slopes.csv]. A retroauricular electrode therefore loses signal with distance more than three times as fast as a jaw electrode does, which is the geometric mechanism behind the labial-group deficit reported in ğ4.3. Lines are least-squares fits over the overlapping range only, so the comparison is made where both montages have data rather than extrapolated across the ear's longer reach.

Limb studies cannot make this separation, because adding a fat layer changes material properties and source-to-electrode distance together (Kuiken et al. 2003). A labelled head model can, because conductivity is changed with geometry held exactly fixed. This is a comparison the limb geometry structurally cannot support, and it is available here for the cost of one additional solve.

The material share is not a dose response. Across the muscles for which a layer profile exists, it correlates *negatively* with the adipose fraction of the muscle-to-skin path (Spearman = −0.955, p = 0.001, n = 7): platysma sits at 0.650 fat fraction and 0.6 per cent material share, while sternocleidomastoid sits at 0.225 and 21 per cent. At n = 7 a single muscle could carry that relationship, so it was tested: dropping each muscle in turn leaves between −0.928 and −0.986 and p below 0.01 in all seven cases [`results/04t_correlation_robustness.csv`]; the seven muscles and that relationship are plotted in Figure 6. The strength of that relationship shows the swap is measuring adipose path rather than something incidental, but its sign shows it does so through cancellation. The reported share is |gap| / |gap|, and a muscle embedded uniformly in fat has both the jaw and the ear route shifted together, so the change subtracts out of the ratio. The quantity that would track positively is the *difference* between how the two routes traverse fat, which this study does not form. This is the same cancellation that makes a uniform magnitude offset invisible in every ratio reported here, arriving in a place where it was not anticipated.

Applied to the three muscles whose gaps come closest to zero, temporalis, sternocleidomastoid and lateral pterygoid, the same decomposition returns a modest term whose sign varies between them.

For temporalis the contrast acts against the gap rather than for it: removing it would enlarge the gap from −3.801 to −4.923 dB. The direction is worth stating because it is the opposite of the intuitive reading, the tissue contrast is not what produces temporalis's proximity to zero; it is part of what holds it there. It does not change the verdict, which is set by the matched−count interval and not by this term. For sternocleidomastoid and lateral pterygoid the contrast acts with the gap, supplying 21.0 and 17.4 per cent of it respectively.

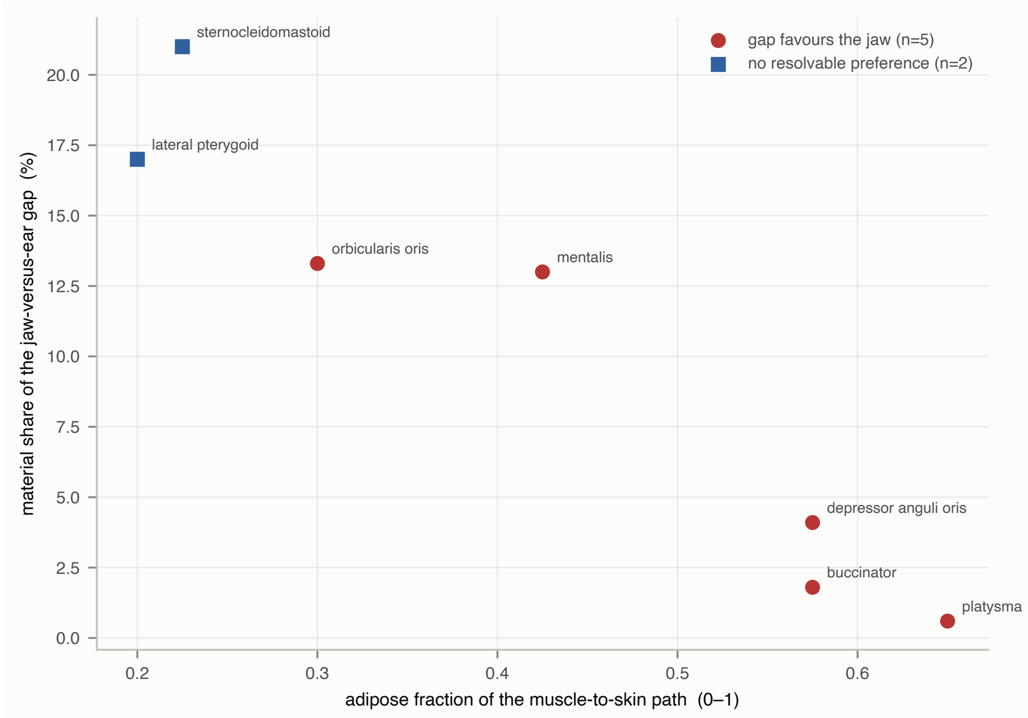

Figure 6: Material share falls as the adipose path grows. For each of the seven muscles with a layer profile, the adipose fraction of the muscle-to-skin path against the material share of that muscle's jaw-versus-ear gap. The relationship is negative (Spearman rho = -0.955, p = 0.001, n = 7): more fat on the path means a *smaller* material share, not a larger one, because a muscle embedded uniformly in fat has both the jaw and the ear route shifted together and the change cancels in the ratio. Leave-one-out leaves rho between -0.928 and -0.986 with p below 0.01 in every case, so no single muscle carries it. Points are marked by whether the muscle's gap resolves to the jaw or reaches no resolvable preference. No trend line is drawn: these are seven independent muscles rather than a series, and Spearman is a rank statistic that claims no functional form.

That distinction predicts which results should transfer between subjects. A margin carried by skeletal attachment geometry inherits only anatomical variance; a margin carried by a tissue-conductivity contrast also inherits variance in adipose distribution, which is the larger and more variable of the two between individuals. The two classes are therefore defined by material share rather than by which montage wins. A margin with a small share should transfer more readily than one with a large share: 0.6 to 13.3 per cent across the labial group, against 21 and 17 per cent for sternocleidomastoid and lateral pterygoid. Temporalis, masseter and medial pterygoid are not classified, because no layer profile exists for them and their share is unmeasured. The prediction is a consequence of the decomposition rather than a separate claim, and it is testable in any second anatomy.

4.4 What this licenses for device design

The design statement this supports is a negative one with a number attached. A retroauricular montage loses the labial group by 8.1 to 22.4 dB, and buys nothing reliable in return. No articulator in this study favours the retroauricular montage once source orientation and electrode count are controlled.

That is more useful to a device team than a small positive margin would have been. An ear-worn form factor is chosen for wearability, not for signal, and the question a designer needs answered is what it costs. The answer is that it costs most of the anterior articulators outright and returns nothing measurable, not that it trades one muscle group for another.

Three apparent advantages did not survive. Temporalis, sternocleidomastoid and lateral pterygoid each showed a retroauricular advantage at the unmatched comparison, and each dissolved: their gaps depend on which four electrodes a device carries rather than on the anatomy (ğ3.1). A device built around any of them would be built on a site-selection lottery. Anyone reading a retroauricular advantage off a model that does not match electrode counts between montages should expect the same.

What does transfer is that placement method matters more than the marginal advantages did. A four-site cluster chosen by anatomical target outperforms an arbitrary four-site draw from the same candidates (ğ3.6). For a device constrained to a few contacts, where they go is the lever that remains.

4.5 Contamination, described muscle by muscle

The EEG literature has documented mastoid and retroauricular electromyographic contamination for decades as a nuisance to be suppressed [Goncharova et al. 2003; Yao et al. 2019]. This model says what that contamination consists of. The three compartments a retroauricular electrode couples to most strongly, relative to the canonical jaw montage, and whose sensitivity fields are shown in Figure 7, are temporalis, sternocleidomastoid and lateral pterygoid, and their best positions differ, so contamination at `cg01` is not the same mixture as contamination at `cg08`.

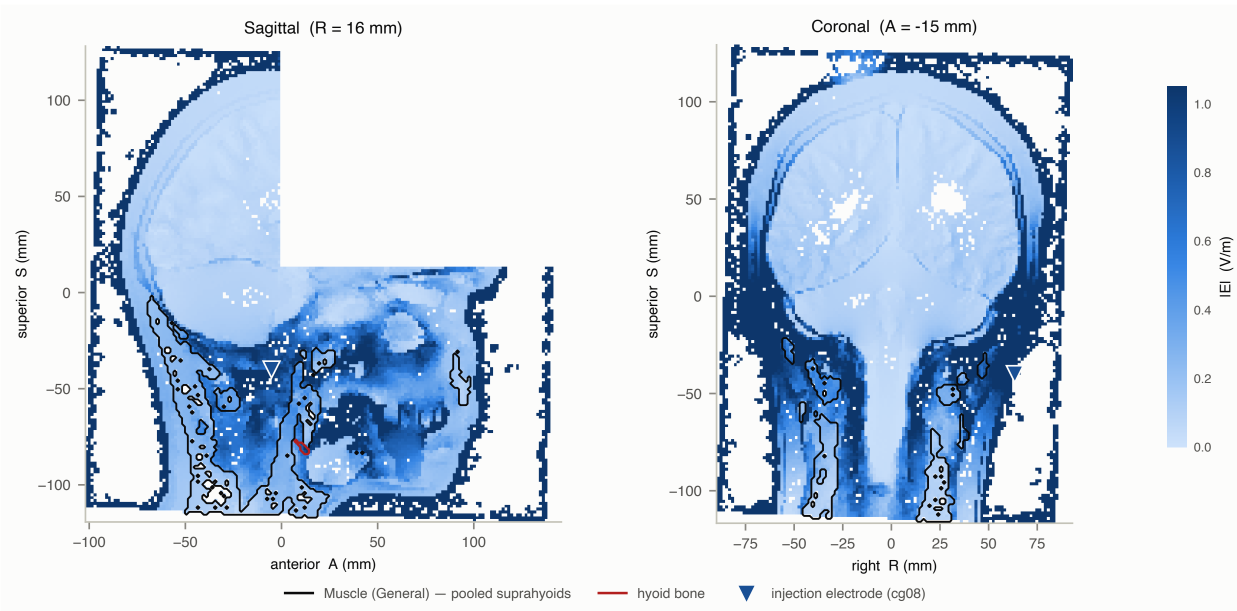

Figure 7: Suprahyoid sensitivity field. Sagittal and coronal slices of |E| through the pooled Muscle (General) compartment for the retroauricular montage, with the mastoid notch, hyoid greater horn and the corridor between them overlaid. Reported as a field rather than as per-muscle values because MIDA does not segment the suprahyoid group.

That is usable in the rejection direction as well as the sensing one. A spatial filter informed by which muscle dominates at which contact is a different object from one treating retroauricular electromyography as a single nuisance component.

4.6 A specific prediction for a companion experiment

This model makes a falsifiable prediction for a physical experiment recording both montages simultaneously, and the prediction is a null.

An eight-channel rig split four jaw and four retroauricular, recording identical utterances, should show **no articulator for which the retroauricular channels carry more information than the jaw channels**, and should show the labial group degrading sharply at the ear. A result finding a reliable retroauricular advantage for temporalis, sternocleidomastoid or lateral pterygoid would **falsify this model**, and would most likely indicate that a real muscle's fibre geometry differs from the field derived here, or that mechanical or acoustic coupling contributes signal this electrical model does not represent.

Stating the direction in advance matters, because the alternative reading is available and would be unfalsifiable. Had the prediction been "the ear retains temporalis-driven gestures", an experiment finding no advantage could be explained as insufficient sensitivity rather than as evidence against the model. As stated, the null is the prediction and a positive finding is the refutation.

4.8 How the retroauricular advantage dissolved

An apparent retroauricular advantage was present at every intermediate stage of this analysis and survived to the point of being written into a draft. It did not survive the controls, and the way it disappeared is worth reporting because each control is individually standard and none was applied in response to the result.

Stage	Temporalis gap
field magnitude, best of 14 ear sites	−3.92 dB
projected onto source orientation	−3.31 dB
matched electrode counts, four sites each	−2.57 dB, interval [−3.31, −0.03]
derived per-voxel fibre field	**−1.15 dB, interval [−1.45, +5.46] spans zero**

Three corrections, each motivated by a different defect, each moving the estimate the same way. Reporting the field magnitude rather than the lead field projected onto a source orientation overstates coupling, because the magnitude is the maximum over orientations. Comparing the best of fourteen candidate sites against the best of four rewards electrode density rather than placement. And assuming a fibre direction rather than deriving one from the label volume permitted a claim the derived field does not support.

None of these is exotic. Each is the kind of simplification a forward-modelling study makes for defensible reasons, and each individually shifts the estimate by around a decibel. Their product is the difference between a 3.92 dB advantage and none.

Two features of this sequence are worth separating, because only one of them is evidence. The **monotone** drift, every correction moving the same way, is suggestive but not probative; corrections that each remove an optimistic assumption will tend to move one way by construction. What is probative is that the **final** step, the one that crosses zero, removes an assumption rather than adding one, and was pre-committed (ğ2.8) before the derived field existed. An effect that survives only under an assumption its own anatomy can replace is not an effect.

We report this because the intermediate results were not obviously wrong. Each was internally consistent, cleared its measurement floor, and reproduced an a-priori anatomical prediction, the three muscles that appeared to favour the ear are the three whose attachments sit at or near the temporal bone, which is exactly what one would predict and exactly what makes a spurious result convincing.

4.7 Limitations

Ten of eighteen muscles are modelled, and the two carrying the strongest version of the anatomical argument are not among them. MIDA does not individually segment the suprahyoid group or the tongue. Posterior digastric and stylohyoid, the two muscles that anchor at the mastoid notch and styloid process, and that motivated the retroauricular hypothesis in the first place — are therefore absent from the per-muscle comparison. The model is silent exactly where the a-priori argument was strongest, the muscles whose attachments most directly motivated a retroauricular montage are the ones it cannot test, and the spatial sensitivity field reported over the pooled compartments is a partial substitute rather than an equivalent one.

MIDA is a single subject, and between-subject variance in muscle geometry, adipose thickness and pinna position cannot be estimated from one head. The adipose decomposition narrows what that means, but unevenly, and the unevenness is itself informative. The labial group and temporalis are carried by geometry, which is comparatively conserved between individuals; sternocleidomastoid and lateral pterygoid draw 21.0 and 17.4 per cent of their gap from the conductivity contrast, so of the three muscles nearest zero they are the two whose position is most dependent on tissue properties rather than on geometry, and the two whose gaps should be expected to move most with subject adiposity. This is a reason to expect differential generalisation, not a demonstration of any of it. Only a second anatomy demonstrates that.

The inferior boundary is an unquantified limitation whose bias runs against the ear. A neck-extended mesh was built specifically to measure it and did not conserve charge, so the pre-committed decision rule was recorded unexecuted rather than applied or revised. What can be said is ğ3.4: excluding the two near-cut jaw sites moves the median gap by only −0.17 dB and flips no signs, and seven of ten muscles are immune by construction. The magnitude of the residual bias is unknown rather than estimated, and its direction flatters this paper's own headline comparison.

Static geometry. The model is solved on a single anatomical configuration. Articulation moves the tongue, opens and closes the oral cavity and alters the airway, all of which change the volume conductor during the task being measured. Nothing here quantifies that, and the jaw montage sits closer to the moving structures than the ear montage does.

Quasi-static assumption, standard at surface electromyography frequencies and stated for completeness.

Fibre orientation is bounded, not known (ğ2.5). A fibre tensor reaches two of ten segmented muscles, and rows without one are reported as not applied rather than as zero change.

The orientation constraint is derived for one muscle and assumed for the rest, and it cannot be extended to all of them. The anatomically-constrained sweep of ğ2.5.1 derives a per-voxel fibre field for temporalis by pointing each voxel at its mandibular insertion. The other nine segmented articulators are swept uniformly over the hemisphere, which assumes source orientation is uniformly distributed for each. That assumption is stated once and inherited throughout, and it is the assumption the temporalis derivation exists to remove.

The construction requires a discrete bony insertion for voxels to point at, so the remainder do not form a single pending set. Sternocleidomastoid, medial pterygoid and mentalis are strap-like with fibres along the compartment long axis, and admit it directly. Masseter, lateral pterygoid and depressor anguli oris are multi-part or converging, superficial and deep layers at different angles, two heads, and a converging triangular sheet respectively, and a per-voxel fan toward a common attachment is the appropriate treatment for them, untested here. Orbicularis oris admits it in no form: it is a sphincter whose fibres run in a ring, with no bony insertion, so a principal axis is a category error rather than a missing measurement. Buccinator blends into that sphincter at the modiolus and inherits the same problem. Platysma is a broad sheet.

The limitation is therefore not that nine derivations are outstanding. It is that the treatment which changed the temporalis result is available for some compartments, inappropriate for others, and impossible for at least one.

One jaw site is withheld, and its omission runs against the reported jaw advantage. `throat_scm` carries no coordinate because MIDA's sternocleidomastoid is truncated by the inferior cut plane, which biases its centroid posteriorly, so no defensible automatic placement exists (ğ2.3). It is the jaw site nearest the ear montage, so a jaw montage carrying it would be compared from a position closer to the retroauricular sites than any jaw site actually used. The direction of that omission is toward a smaller jaw advantage than is reported.

Tables

Table 1, Tissue conductivities. All 116 MIDA labels with assigned conductivity, source (SimNIBS 4.6 default / IT'IS LF v4.2 / judgement), frequency, plausible range for judgement rows, volume fraction and minimum distance to the nearest electrode. Sorted by volume fraction × proximity. [`results/table1_conductivities.csv`]

Table 2, Tissue layer stack beneath each canonical site. Millimetres per MIDA tissue along the ray from each electrode through the full thickness of its target. *Target thickness traversed* is summed from `results/02_layer_profile.csv` over the target label. *Fat before target* is derived from `results/02_path_composition.csv` as the adipose percentage of the electrode-to-target path multiplied by that path length, i.e. adipose encountered **before** reaching the target, not over the whole ray.

Site	Target	Target thickness traversed	Fat before target
`submaxillary`	Mandible	27.25 mm	2.85 mm
`pre_tragus`	Masseter	17.25 mm	5.68 mm
`midjaw`	Masseter	16.50 mm	9.05 mm
`submental_lat`	Mandible	11.00 mm	1.50 mm
`buccal`	Buccinator	8.00 mm	8.62 mm
`submental_mid`	Mandible	7.25 mm	7.04 mm
`hyoid`	Hyoid Bone	6.25 mm	14.01 mm
`above_ear`	Temporalis	6.00 mm	2.00 mm
`mental`	Mentalis	2.75 mm	4.87 mm

Table 3, Error budget. Read the last two columns first: every published claim here is a ratio, so a term that scales the whole lead field equally cancels and never reaches a conclusion, while a term varying between sites survives into all of them.

#	Term	What sets it	Affects absolute	Affects ratios	Value
1	Discretisation	finite element size	yes	partly — directional, unquantified	not separable from term 6 at current precision
2	Interface proximity	source near a conductivity boundary	yes	yes — directional, unquantified	requires a geometry decoupling eccentricity from interface distance; not measured

#	Term	What sets it	Affects absolute	Affects ratios	Value
3	Inferior boundary	MIDA's cut face	yes	yes — jaw sites, not ear; directional, unquantified	unquantified; direction known, magnitude not bounded
4	Muscle anisotropy	tensor vs scalar	**yes — ~5 dB in medial pterygoid, ~4.5 dB in SCM**	**no — below the floor**	statistic A: largest change to any gap is **−0.085 dB** (SCM); medial pterygoid −0.010, temporalis +0.137, lateral pterygoid +0.036, all under the 0.27 dB floor. The absolute lead field IS affected; the reclassification is specific to ratios. Tensor on 2 of 10 compartments, the rest NOT APPLIED
5	Fibre orientation	n unknown in MIDA	yes	yes	per-muscle min–max envelope
6	Electrode meshing	contact area from incidental surface triangulation	yes	**yes — per-site**	**0.27 dB, 95 % CI [0.17, 0.65], n = 6**
7	Single anatomy	MIDA is one subject	yes	unknown — directional, unquantified	not quantifiable from one head
8	Delivered current	injected vs requested per solve	yes	**no — corrected, not bounded**	0.887–1.075 × requested across 22 solves, measured per solve by the tet-patch integral. Each site's lead field is divided by its own delivered current (ğ2.4), so the term does not enter any reported ratio. It is listed here because it was measured and corrected, not because it remains an uncertainty: the 1.67 dB spread it would otherwise contribute is six times the row-6 floor and could not have been bounded by it

#	Term	What sets it	Affects absolute	Affects ratios	Value
9	Adipose conductivity	fat at 0.025 vs muscle 0.355 S/m	yes	**yes — and the SIGN differs by muscle**	Reported **per muscle only** — a population differential across sites has no clean definition under statistic A (the median change over muscles is +0.01 dB and conceals a sign that spans −2.86 to +1.09). Statistic A, per muscle, sign varies: **−1.121 dB** for temporalis (acts against the gap), **+0.411 dB** for sternocleido-mastoid and **+0.323 dB** for lateral pterygoid (act with it), **−2.863 to +1.093 dB** across the labial group. No single figure is admissible — the sign differs by muscle. Shares are not quoted here because they are dominated by the denominator; see ğ4.3.

Row 6 is measured by rotating the electrode array and the source points together on a fixed mesh, which preserves every source-to-electrode vector (verified to 2.8×10^{-14} mm) so the exact answer is identical across draws and the whole spread is realisation noise. The term splits into a common-mode part (SD 4.93 pp, 0.42 dB, cancels in a ratio) and an electrode-specific residual (SD 3.18 pp, 0.27 dB, does not). The reported statistic is the mean over 16 electrodes of the per-electrode SD across 6 draws, with a chi-square interval at df = 5.

Row 7 is deliberately left unquantified. A single-subject model cannot estimate its own between-subject variance.

Table 4, Which montage sees which muscle, on two robustness axes. Gap in dB between the jaw and retroauricular montages, positive favouring the jaw, taken as the median over 200 source orientations of the per-orientation gap (statistic A) against the four pre-registered retroauricular sites. Electrode counts are matched at 4 per montage. Because 5 jaw sites are admissible and the comparison takes 4, every value is reported as the envelope over all 5 admissible subsets rather than for one chosen subset. Rows marked unstable change verdict between subsets and are described in the text. *Site-robust* asks whether a random draw of four of the fourteen ear sites still excludes zero. *Orientation agreement* is the fraction of sampled orientations agreeing with the median verdict.

Muscle	Gap (dB), envelope over subsets	Site-robust	Orientation agreement	Verdict
mentalis	+20.98 to +22.40	yes, all 5	100.0 %	**jaw, robust on both axes**
depressor anguli oris	+14.70 to +15.98	yes, all 5	100.0 %	**jaw, robust on both axes**
buccinator	+9.45 to +10.39	yes, all 5	100.0 %	**jaw, robust on both axes**

Muscle	Gap (dB), envelope over subsets	Site-robust	Orientation agreement	Verdict
platysma	+9.90 to +10.27	yes, all 5	100.0 %	**jaw, robust on both axes**
orbicularis oris	+8.11 to +9.02	yes, all 5	100.0 %	**jaw, robust on both axes**
masseter	+1.20 to +2.22	yes, all 5	56.0–68.5 %	**jaw, site-robust but orientation-dependent**
medial pterygoid	+0.86 to +1.33	yes, all 5	59.5–66.5 %	**jaw, site-robust but orientation-dependent**
sternocleidomastoid	-2.53 to -0.97	1 of 5	60.5–65.0 %	**unstable across subsets**
lateral pterygoid	-3.61 to -1.53	1 of 5	65.5–72.0 %	**unstable across subsets**
temporalis	-5.42 to -2.57	yes, all 5	92.0–100.0 %	**ear, robust on both axes**

No jaw site is preferred by the design over any other, so rather than select one 4-site subset the comparison is run over all 5 admissible subsets and reported as an envelope. 5 of 10 muscles favour the jaw montage on both robustness axes in every subset, spanning 8.11 to 22.40 dB. No muscle favours the retroauricular montage on both axes in any subset.

8 muscles return an identical verdict in all 5 subsets. Two do not. Lateral pterygoid gives no resolvable preference across four subsets (−1.53 to −1.59 dB) and −3.61 dB in the fifth; sternocleidomastoid gives −0.97 to −1.44 dB across four and −2.53 dB in the fifth. In both cases the deviating subset is the one dropping `midjaw`, and in both cases the deviating value is site-robust but orientation-dependent, with agreement of 72.0 and 65.0 per cent against the 90 per cent required for a robust preference. Neither reaches a retroauricular preference on both axes, and neither is among the 5 muscles carrying the jaw advantage. The subset that deviates is the one dropping `midjaw`, which has the largest perpendicular clearance to the cut plane of any jaw site at 63.76 mm; the instability follows from removing the site furthest from the truncation, not one near it.

Temporalis is reported here under the uniform orientation sweep, for comparability with the other nine muscles. Under the fibre field derived from the label volume (ğ2.5.1) it reads −1.15 dB with a random-4 interval of [−1.45, +5.46], which crosses zero, and that is the value the paper's conclusions use (ğ3.1, ğ4.1). The two treatments disagree; the derived one governs because it removes an assumption rather than adding one. This row shows what the assumption-free treatment gives, so the dependence is visible rather than buried.

Supplementary Figure S1. Air-void inventory: volume and depth below skin for all nine air-filled compartments in MIDA.

Data and code availability

All analysis code, electrode definitions, conductivity assignments and result tables are available at `github.com/Somach-Systems-Inc/ear-emg-forward-model`. The MIDA model itself cannot be redistributed under its licence and must be obtained from the IT'IS Foundation; `REPRODUCTION.md` documents the one manual step. Commits `fa583f6` (2026-08-02) and the lead-field commit (2026-08-03) are cited in ğ2.8 as the pre-registration record.

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